

1. Sum of First N Natural Numbers

SUM OF FIRST N NATURAL NUMBERS

$N = 5$

$$1 + 2 + 3 + 4 + 5 = 15$$

$N = 100$

$$1 + 2 + 3 + 4 + 5 \dots + 100 = 5050$$

$N = 1000$

$$1 + 2 + 3 + 4 + 5 \dots + 1000 =$$

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Untitled Notebook 72

Introduction to Problem Solving (3)-1.pdf

$N = 100$

$$1 + 2 + 3 + 4 + 5 \dots + 100 = 5050$$

$N = 1000$

$$1 + 2 + 3 + 4 + 5 \dots + 1000 = 505000$$

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Paste Insert Text Insert Image

$$S = 1 + 2 + 3 + 4 + 5 \dots + 100$$
$$S = 100 + 99 + 98 + 97 \dots + 1$$

101 101 101 101 101

$$\begin{aligned}
 S &= 1 + 2 + 3 + 4 + 5 \dots + 100 \\
 S &= 100 + 99 + 98 + 97 \dots + 1 \\
 + \\
 \hline
 2S &= (101) + (101) + (101) \dots + (101) \\
 2S &= 101 \times 100 \\
 S &= \frac{101 \times 100}{2} = 101 \times 50 = \underline{\underline{5050}}
 \end{aligned}$$

$$\begin{aligned}
 S &= 1 + 2 + 3 + 4 \dots + N \\
 S &= N + (N-1) + (N-2) \dots + 1 \\
 + \\
 \hline
 2S &= (N+1) + (N+1) + (N+1) \dots + (N+1) \\
 2S &= (N+1) \times N \\
 S &= \frac{(N+1) \times N}{2} \\
 \boxed{S &= \frac{N(N+1)}{2}} \star
 \end{aligned}$$

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2. Find how many terms are there in the sequence? (Means number in sequence)

The screenshot shows a digital notebook interface with two tabs: "Untitled Notebook 72" and "Introduction to Problem Solving (3)-1.pdf". The notebook page contains the sequence $3, 4, 5, 6, 7, 8, \dots, 100$ written in white. A red underline is drawn under the sequence. Below the sequence, the text "How many terms are there in the above sequence?" is written in white. The interface includes a toolbar with various drawing and editing tools.

The screenshot shows the same digital notebook interface as the previous one, but with the solution to the problem. The sequence $3, 4, 5, 6, 7, 8, \dots, 100$ is written in white, followed by $\Rightarrow 98$ in yellow. Below this, the text "How many terms are there in the above sequence?" is written in white. The solution is shown as $3, 4, 5, 6, 7, 8 \Rightarrow 8 - 3 + 1 = 6$ in white. At the bottom, the general formula is given as $L, L+1, L+2, L+3, \dots, R \Rightarrow (R - L + 1) \star$ in white, with the expression $(R - L + 1)$ circled in yellow and a yellow star next to it. The interface includes a toolbar with various drawing and editing tools.

3. Inclusive and Exclusive

Handwritten notes explaining inclusive and exclusive ranges:

- $[\quad] \rightarrow$ Inclusion
- $(\quad) \rightarrow$ Exclusion
- $[3, 8] \rightarrow 3, 4, 5, 6, 7, 8$
- $[3, 8) \rightarrow 3, 4, 5, 6, 7$
- $(3, 8] \rightarrow 4, 5, 6, 7, 8$
- $(3, 8) \rightarrow 4, 5, 6, 7$

★ $[L, R] \Rightarrow R - L + 1$

$[L, R) \Rightarrow R - L$

$(L, R] \Rightarrow R - L$

$(L, R) \Rightarrow R - L - 1$

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4. Count Element in range

Handwritten notes on counting elements in a range:

COUNT ELEMENTS IN A RANGE

$1 + 2 + 3 + 4 + 5 \dots + N = \frac{N(N+1)}{2}$

$[L, R] \Rightarrow R - L + 1$

5. Factors

FACTORS

12 $\rightarrow$ 1, 2, 3, 4, 6, 12 $\Rightarrow$ 6 Factors

15 $\rightarrow$ 1, 3, 5, 15 $\Rightarrow$ 4 factors N

count Factors of N

Min. Factor 1

Max. Factor

count Factors of N

Min. Factor 1

Max. Factor N

```
count = 0;
for (i = 1; i <= N; i++)
{
    if (i is a factor of N)
    {
        count++;
    }
}
```

print(count);

1
Count $\leq$
S.

12 $\rightarrow$ 1, 2, 3, 4, 6, 12 $\Rightarrow$ 6 Factors

15 $\rightarrow$ 1, 3, 5, 15 $\Rightarrow$ 4 factors N

count Factors of N

Min. Factor 1

Max. Factor N

```
count = 0;
for (i = 1; i <= N; i++)
{
    if (i is a factor of N)
    {
        count++;
    }
}
```

print(count);

$N \% i == 0$

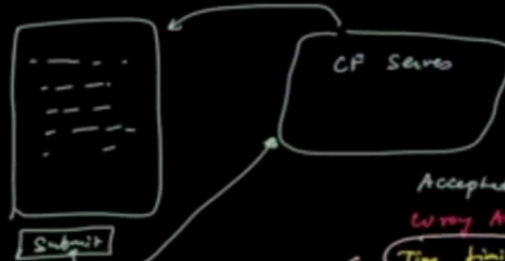
print(count);

#iterations $\Rightarrow$ How many times the loop runs?

CF

Time limit = 1 sec

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Code is taking too long to execute.

Accepted

Wrong Answer

Time limit Exceeded

Runtime Error

int Arr[10];
for (i=1; i<=9; i++)

Arr[i] = i;

Arr[10] = 10;

Arr[100] = 100;

Assumption
Standard

1 sec $\rightarrow 10^8$ iterations

2 sec $\rightarrow 2 \times 10^8$ iterations

10 sec $\rightarrow 10 \times 10^8$ iterations = 10^9 iterations

Arr[8] = 8;

Arr[10] = 10;

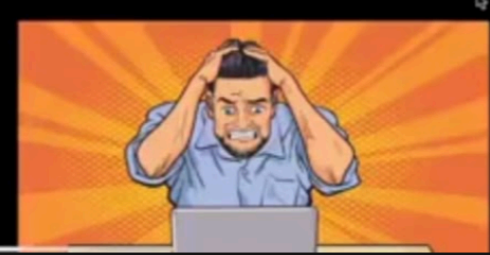
Arr[100] = 100;

Introduction to Problem Solving

<u>N</u>	<u># iters</u>	<u>Time</u>
10^8	10^8	1 sec
10^9	10^9	10 sec $\times \left(\frac{10^9}{10^8}\right)$
10^{12}	10^{12}	10^4 seconds $\left(\frac{10^{12}}{10^8}\right)$
$\vdots$		10,000
10^{15}	10^{15}	10^{10} sec $\left(\frac{10^{15}}{10^8}\right)$
		300+ years ↓ - Minutes - Hours - Days 1 4 Years

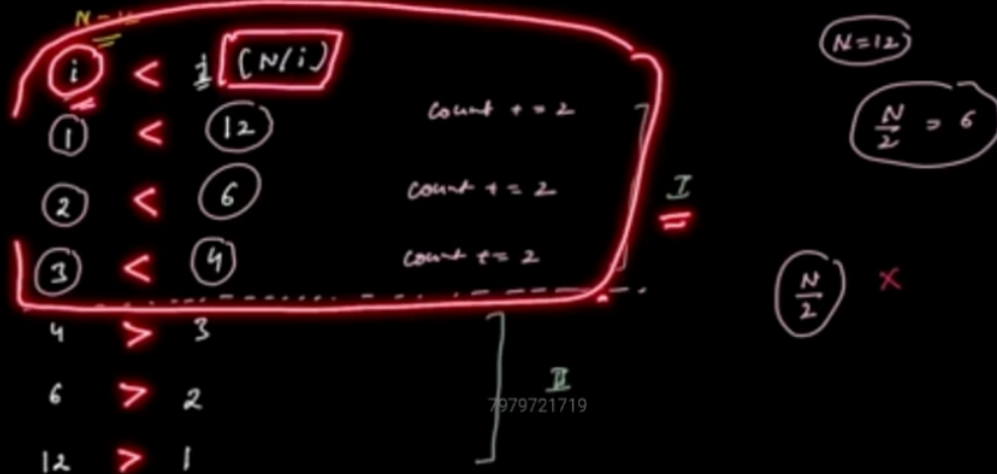
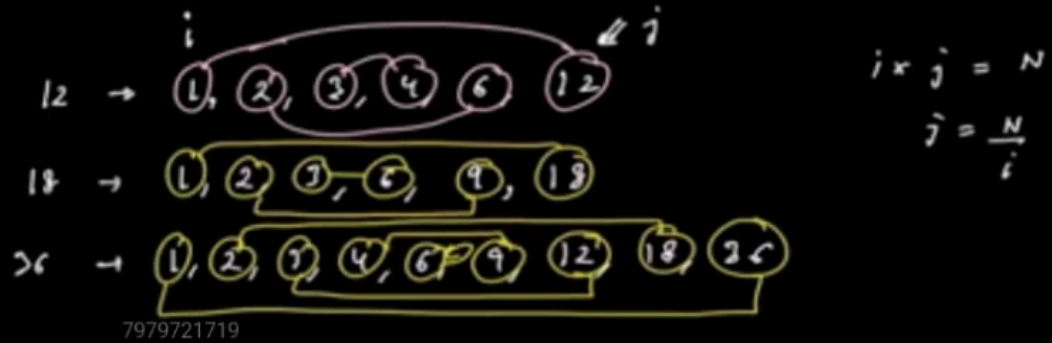
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WHAT COULD HAPPEN IN 300 YEARS ?

Let's optimize now

OPTIMISING THE SOLUTION



$$i < \frac{N}{i} \Rightarrow i^2 < N \Rightarrow \boxed{i < \sqrt{N}}$$

Perfect Square

Perfect Square

$$N = 36$$

i
①

$\frac{N}{i}$ (N/i)
③⑥

Count += 2

②

①⑧

"

③

①②

"

④

⑨

"

⑥

6

Count += 1

9

4

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12

3

18

2

36

1

```
9:47 AM Tue 9 Jan
Untitled Notebook 22
Introduction to Problem Solving (D-1).pdf
133%
Count = 0
for (i = 1; i ≤ √N; i++)
{
    if (N % i == 0)
    {
        j = N / i;
        if (i == j)
        {
            count++;
        }
        else
        {
            count += 2;
        }
    }
}
print (count);
```

```
Count = 0
for (i = 1; i ≤ √N; i++)
{
    if (N % i == 0)
    {
        j = N / i;
        if (i == j)
        {
            count++;
        }
        else
        {
            count += 2;
        }
    }
}
print (count);
```

$N = 10^6$

$\sqrt{N}$

$\sqrt{10^6} = 10^3$

$(\sqrt{N})$

<u>N</u>	<u># iterations</u>	<u>Time</u>	$10^8 \text{ iters} \rightarrow 1 \text{ sec}$
10^8	10^4	< 1 second	
10^{12}	10^6	< 1 second	
10^{16}	10^8	1 second	
10^{18}	10^9	10 seconds	

300 + years

↓

10 seconds

6.

①	for (i = 1; i <= N; i++)	①
	{	
	==	
	}	
②	for (i = N; i >= 1; i--)	①
	{	
	==	
	}	
③	for (i = 1; i <= N; i++)	①
	{	
	==	
	}	

7. Time Complexity

COUNT ITERATIONS

```
int i = N;
while(i > 1)
{
    i = i / 2;
}
```

$N = 16$ 4 times

$i = 16$
↓
8
↓
4
↓
2
↓
1
==

$N = 72$ 5 times

$i = 72$
↓
36
↓
18
↓
9
↓
4
↓
2
↓
1
==

$N ??$
 $\sqrt{N} ??$

```
int i = N;
while(i > 1)
{
    i = i / 2;
}
```

$N = 4096$ 12 times $2^{12} = 4096$

$i = 4096$
↓
2048
↓
1024
↓
512
↓
256
↓
128
↓
64
↓
32
↓
16
↓
8
↓
4
↓
2
↓
1

$2^k = N$
 $k = \log_2 N$

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COUNT ITERATIONS

```
int i = 1;
while(i < N)
{
    i = i * 2;
}
```

$\log_2 N$

$N = 16$

$i = 1$
↓
2
↓
4
↓
8
↓
16

COUNT ITERATIONS

```
int i = 0;
while(i <= N)
{
    i = i * 2;
}
```

Infinite

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COUNT ITERATIONS

```
for(i = 1; i <= N; i++)
{
    for(j = 1; j <= N; j++)
    {
        print(i + j);
    }
}
```

<u>i</u>	<u>j</u>	<u># iters</u>
1	1 - N	N
2	"	"
3	"	"
⋮	⋮	⋮
N	"	"

$$N \times N = N^2$$

COUNT ITERATIONS

```
for(i = 1; i <= N; i++)
{
    for(j = 1; j <= i; j++)
    {
        print(i + j);
    }
}
```

<u>i</u>	<u>j</u>	<u># iters</u>
1	1 - 1	1
2	1 - 2	2
3	1 - 3	3
⋮	⋮	⋮
N	1 - N	N

$$\frac{N(N+1)}{2}$$

8.

COMPARING IDEAS / APPROACHES

Sort an array

Idea 1

12 seconds
(Windows XP)
↓
8 seconds
(Rust)
↓
6 seconds (Mount Everest)

Volcano (External Factor)

Idea 2

10 seconds
(Mac OS)
↓
6 seconds (Rust)
(Mount Everest)

Jang

Execution Time is not a good parameter to compare two ideas / algorithms.

Number of iteration is a better parameter to compare 2 ideas.

COMPARING IDEAS / APPROACHES

Sort an array

Idea 1

12 seconds
(Windows XP)
↓
8 seconds
(Rust)
↓
6 seconds (Mount Everest)

Volcano (External Factor)

Idea 2

10 seconds
(Mac OS)
↓
6 seconds (Rust)
(Mount Everest)

Jang

Execution Time is not a good parameter to compare two ideas / algorithms.

BIG O TIME COMPLEXITY

- 1) Find # of iterations
- 2) Ignore / Remove all the constants
- 3) Remove all lower contribution terms / lower ordered terms

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$$\left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \right] \rightarrow \cancel{10} N^2 + \cancel{5} N + \cancel{600}$$

$$\quad \quad \quad \downarrow$$

$$\quad \quad \quad \textcircled{N^2} + N$$



$$\left[\begin{array}{c} \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \\ \text{---} \end{array} \right] \rightarrow \cancel{10} N^2 + \cancel{5} N + \cancel{600}$$

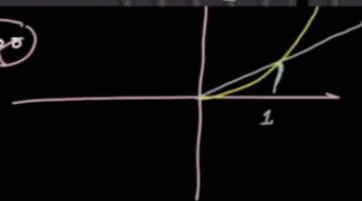
$$\quad \quad \quad \downarrow$$

$$\quad \quad \quad \textcircled{N^2} + \cancel{N}$$

$$\quad \quad \quad \downarrow$$

$$\quad \quad \quad \textcircled{N^2}$$

$$\quad \quad \quad \underline{\underline{O(N^2)}}$$



$$\underline{N^3} + \cancel{N} + \underline{N^3 \cdot \log N} + \cancel{N^2 \cdot \sqrt{N}} \quad N^{2 + \frac{1}{2} = \frac{5}{2}}$$

$$O(N^3 \cdot \log(N))$$

3
 $\frac{1000}{3} \Rightarrow \boxed{O(1)}$

- 0) $N!$
- 1) 2^N
- 2) N^2
- 3) N
- 4) $N \log N$
- 5) $N \sqrt{N}$
- 6) $\log_2 N$
- 7) $\sqrt{N}$
- 8) $O(1)$

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$$O(1) < O(\log N) < O(\sqrt{N}) < O(N) < O(N \log N) \\
< O(N \sqrt{N}) < O(N^2) < O(2^N) < O(N!)$$